Rishi Raj

Doctoral Researcher in Mathematical Physics • Aspiring Machine Learning Engineer

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Versatile and results-driven Machine Learning Engineer with a strong foundation in web application development and applied machine learning. Proficient in building scalable AI-powered applications using modern frameworks like Node.js, React, and TypeScript, and integrating robust machine learning workflows with Python.

Expert in deploying and optimizing AI solutions, with hands-on experience in large language models (LLMs), natural language processing (NLP), and cloud-based model deployments. Skilled in MLOps/DevOps best practices, ensuring efficient, reliable, and high-performance AI pipelines.

Collaborative and innovation-focused professional, adept at working with cross-functional teams to solve real-world challenges. Committed to staying updated with advancements in AI and applying cutting-edge technologies to deliver transformative solutions.

TECHNICAL SKILLS

Programming Languages: Proficient in Python (scientific and production-grade), C/C++ (high-performance computing) with CMake, JavaScript / TypeScript (full-stack systems)
ML & AI Frameworks: PyTorch, scikit-learn, XGBoost, Hugging Face, Transformers, Weights & Biases (wandb), Pandas visualization with Seaborn, *R*, ROOT
Cloud & DevOps: End-to-end model deployment pipelines with AWS & GCP, Docker, CI/CD via GitHub Actions, ML model deployment & monitoring, MLOps best practices
Scientific Computing: Mathematica, GNU Scientific Library (GSL), Boost C++, Julia, SageMath
Dev Tools & Version Control: Advanced Git workflows, Linux System Administration, VS Code
Full Stack Web Development: Node.js, Flask, FastAPI, React based frameworks like Next.js, MongoDB, SQL
Core Competencies: Research-grade problem solving, Continuous Learning, Collaborative Problem-Solving, Cross-Functional Communication, Leadership, Adaptability
Langauges: English (native), French (beginner), German (beginner, actively pursuing a langauge course)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant May 2023 – Jul 2023
Supervised by Boris Pioline @ LPTHE, Sorbonne Université, Paris, France

- Utilized Mathematica and SageMath to study bound states of BPS particles in the Coulomb branch of $\mathcal{N} = 4$ supersymmetric quiver quantum mechanics (QQM), revealing structural features of the moduli space.
- Performed a comparative analysis of BPS index computation via attractor flow trees, Coulomb branch localization, and modular anomaly methods, offering insights into wall-crossing behavior.
- Helped clarify the contribution of scaling solutions using the Minimum Modification Hypothesis (MMH), reproducing attractor flow tree results from the Coulomb branch formula.
- Applied graph-theoretic and numerical methods to compute physical invariants, improving precision and performance in multi-centered black hole configurations.

Web Development Freelancer Nov 2021 – Jul 2022
Zenoholics Pvt. Ltd., Bangalore, India

- Lead the modernization of full-stack applications using React.js, Next.js, and MongoDB, achieving a 30% performance gain through architectural refinement and performance profiling.
- Leveraged advanced diagnostics tools—DevTools, Profiler API, Lighthouse, Postman, and Bundle Analyzer—to identify bottlenecks and guide optimization across the stack.
- Engineered robust CI/CD workflows with GitHub Actions, Docker, and AWS; incorporated Kubernetes for container orchestration and scalable deployments.
- Gained hands-on experience in scaling distributed systems via RabbitMQ, connection pooling, caching strategies, lazy loading, CDNs, and load testing, enhancing backend resilience and frontend responsiveness.

EDUCATION

Doctor of Philosophy, PhD in Mathematical Physics Oct 2023 – Sept 2026 (expected)
Sorbonne Université, Paris, France
TODO Research focus on *BPS Black Holes and Donaldson-Thomas Invariants* under the supervision of Boris Pioline. Currently exploring phase space partitioning and attractor flow trees.
Advanced Deep Learning (ongoing), Applied Machine Learning, String Theory, Quantum Field Theory II, Supersymmetry and Supergravity, Differential Geometry

Master of Science, MS in Physics Aug 2022 – Jul 2023
Indian Institute of Technology, IIT Madras, Chennai, India 9.18/10 CGPA
Received the *Mr S. Venkitaramanan, I.A.S Retd* prize and *Electronics For You* prize for securing the highest CGPA during two academic years. Master's thesis on *Black Holes through the lens of Matrix Theory*.
Quantum Field Theory, Advanced General Relativity, Statistical Physics, Numerical Methods and Programming Lab, Conformal Field Theory, Advanced Quantum Computation and Quantum Information

Bachelor of Science, BS in Physics July 2018 – Aug 2022
Indian Institute of Technology, IIT Madras, Chennai, India 9.18/10 CGPA
Graduated with a Minor in Mathematics and consistently maintained a high CGPA. Conducted significant research in theoretical physics through multiple projects and publications.
Mathematical Physics, Dynamical Systems and Chaos, Differential Topology, Real Analysis, Algebra I, Quantum Mechanics, Probability and Stochastic Processes, Functional Analysis

ACADEMIC PROJECTS

Improving smuon searches with Neural Networks Feb 2025 – Present
Ongoing work with Mark Goodsell et al.
- Investigating advanced machine learning architectures (**TabNet**, **Particle Flow Networks**, **XGBoost**) to enhance signal-background discrimination in smuon searches at the LHC.
- Working with Monte Carlo simulated datasets to develop robust preprocessing pipelines and feature representations for high-dimensional collider data.
- Exploring adjacent applications such as **detector unfolding** and **systematics-aware training**, aiming to improve interpretability and generalization to experimental data.
- Benchmarking model performance against traditional cut-based analyses to quantify gains in statistical sensitivity and background rejection.

Multi-centered Black Hole Quantum Mechanics and Generalized Error Functions Jan 2025 – Present
Ongoing work with Boris Pioline
- Trained models on numerical relativity data for asymptotically $\mathrm{AdS}_4/\mathrm{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

BPS Dendroscopy on Local $\mathbf{P}^1 \times \mathbf{P}^1$ Mar 2024 – Dec 2024
Developed neural networks to reconstruct bulk geometry from boundary physics in holographic setups.
- Trained models on numerical relativity data for asymptotically $\mathrm{AdS}_4/\mathrm{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

Learning Holographic Horizons Jan 2023 – Dec 2023
Developed neural networks to reconstruct bulk geometry from boundary physics in holographic setups.
- Trained models on numerical relativity data for asymptotically $\mathrm{AdS}_4/\mathrm{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

Black Holes through the lens of Matrix theory Oct 2022 – April 2023
Designed a simulation suite for classical and quantum simulations of the M-theory matrix model.
- Achieved performance improvements of up to 17x for simulations of black hole dynamics described by random matrix ensembles.
- Studied deviations in thermalization behavior to identify underlying quantum mechanical properties.

Magnetically Coupled Oscillators Nov 2019 – Jan 2020
Created an experimental setup and theoretical model for magnetically coupled pendula in a research fellowship.
- Designed and conducted experiments to capture data on coupled oscillators with magnetic interactions.

- Built a theoretical framework and performed comparisons between experimental and simulated results, ensuring consistency.

Semiholographic Networks Jul 2019 – Nov 2020

Developed scalar field networks coupled with perfect fluids using a semiholographic framework.

- Implemented the theoretical models and analyzed perturbation responses to understand dynamical properties.
- Collaborated with supervisors to produce insights on emergent behaviors in coupled systems.

arXiv-Rename and PIRSA-DL Automation Tools Jan 2021 – Jun 2022

Engineered Python CLI tools for automating workflows in research and lecture management.

- Developed **arXiv-Rename**, a tool that uses arXiv APIs to rename files based on author and title metadata.
- Built **PIRSA-DL**, a utility for scraping and downloading lecture content from the Perimeter Institute archive.

AWARDS

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OTHER WORK EXPERIENCE

Teaching Assistant Jul 2022 – Nov 2022

PH5060 (Physics Lab I), IIT Madras, Chennai, India

- Assisted students with computational physics problems, including chaotic systems and Monte Carlo simulations.
- Conducted viva sessions and evaluated reports to assess students' understanding and progress.

Horizon Club, IIT Madras Sep 2019

Delivered a lecture series titled *Relativity from Symmetries*, introducing group theory concepts and special relativity to physics enthusiasts

- Designed and presented four lectures focusing on symmetry principles, Lorentz transformations, and physical implications of special relativity.
- Engaged and inspired beginning physics enthusiasts to explore advanced theoretical concepts through interactive discussions.

Music Club, IIT Madras Jul 2019 – Nov 2019

Organized and coordinated musical events, including performances for the annual cultural festival, Saarang

- Led the planning and execution of multiple musical events, ensuring seamless coordination among performers and technical teams.
- Enhanced event management and teamwork skills while handling responsibilities for the institute's cultural programs.